Response to the Referee

A Dual-Track Adaptive Projection Algorithm for a Nonlinear Split Feasibility Problem with Mixed Convex–Pseudoconvex Constraints

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Dear Editor, dear Referee,

We thank the referee for the careful reading of our manuscript and for the constructive comments. We have revised the paper substantially. Below, the referee's comments are reproduced in italics, followed by our response and the corresponding changes. Section, equation, table and page numbers refer to the revised manuscript. A version of the manuscript with all changes marked (additions highlighted, deletions struck through) is provided separately.

Summary of the main changes

1. The Introduction and the Related work section were merged into a single, shorter Introduction (Section 1). The manuscript was shortened from 39 to 25 pages.
2. The paper was rewritten throughout in shorter sentences. Redundant motivation, research questions, lists of difficulties and several subsections were removed.
3. The format of the references was unified, and four works on self-adaptive and decomposition methods for the SFP were added. References that are no longer needed after shortening were removed.
4. The appendix was removed. The per-dataset results are now in the main text (Table 3), and the baseline solvers are described in Section 4.2.1.
5. In addition to the referee's comments, we corrected and sharpened several statements, revised the experimental section and added a new solver diagnostic (Section 4.4). These changes are listed in the final section of this letter.

Major issues

Comment M1.

> *Section 1 "Introduction" and Section 2 "Related work" are suggested to be reorganized. Both sections are suggested to be merged as Section 1. Furthermore, the Related work is unnecessary to be revisited in detail. After merging the Introduction and Related work, it is suggested to limit them to 4 pages. […] Please shorten the paper as much as possible. […] Maybe 25 pages is enough for this paper by removing redundant content and shortening long sentences.*

Response. We agree and have followed this suggestion. The Related work section has been removed and its essential content condensed into the Introduction. The revised Introduction (Section 1, pp. 1–3) recalls the CQ method, self-adaptive and decomposition strategies, and the closest nonlinear antecedents (Li et al., Yen et al., Hu et al.) in a few short paragraphs, and it states three contributions. The whole paper was shortened from 39 to 25 pages.
Changes.

- Introduction rewritten and shortened; the former Section 2 (Related work, including its Table 1) removed. [C3–C19]
- Preliminaries reduced to the facts used later, without subsections (Section 2). [C20–C28]

- The former Section 5 (feature-selection framework) condensed into Section 4.1 of the experiments. [C61–C80]
- Removed: the sparse variant DTAP-S and its experiments, the runtime comparison, the section "Further analyses of robustness, fidelity, and ablation" and the "Summary of findings". [C70, C107–C110] [CẦN XÁC NHẬN: lý do bỏ DTAP-S và so sánh thời gian chạy — chỉ do rút gọn, hay còn lý do khác cần nêu?]
- Conclusion reduced to one paragraph (Section 5). [C111–C115]

Comment M2.

> *In order to make the paper more readable, the writing of the current version of the paper should be substantially improved. Many sentences are ambiguous. […] Please describe the problem, algorithms, etc. in short sentences as much as possible. […]*

Response. We have rewritten the manuscript with this request in mind. Long sentences were split, ambiguous phrases were replaced or removed, and descriptive passages that did not support the results were deleted. In particular, the abstract, the Introduction, the Preliminaries, the opening paragraphs of Section 3 and the experimental section were rewritten. The names in parentheses of lemmas, propositions, remarks and corollaries were removed to shorten the text.
Changes. Abstract; [C1] Section 1; [C3–C14] Section 2; [C20–C28] Section 3.1–Section 3.5; [C30–C43] Section 4; [C71–C110] Section 5. [C111]

Comment M3.

> *The format of the references is inconsistent. Many important references are omitted for SFP methods. E.g., I remember the authors of [23], [35] also have some work on SFP with self-adaptive or decomposition methods.*

Response. Thank you for pointing this out. We now cite the following works of the authors of [23] (Z. Li, D. Han, W. Zhang) and [35] (Q. Yang) on self-adaptive and decomposition methods for the (multiple-sets) SFP:

- W. Zhang, D. Han, Z. Li, A self-adaptive projection method for solving the multiple-sets split feasibility problem, *Inverse Problems* 25 (2009) 115001;
- W. Zhang, D. Han, X. Yuan, An efficient simultaneous method for the constrained multiple-sets split feasibility problem, *Comput. Optim. Appl.* 52 (2012) 825–843;
- J. Zhao, Q. Yang, Self-adaptive projection methods for the multiple-sets split feasibility problem, *Inverse Problems* 27 (2011) 035009;
- Y. Yang, Q. Yang, S. Zhang, Modified alternating direction methods for the modified multiple-sets split feasibility problems, *J. Optim. Theory Appl.* 163 (2014) 130–147.

They are discussed in the first paragraph of the Introduction (p. 1). The reference list now uses one format: initials for given names, journal names in italics, volume, issue, pages, year and, where available, the DOI. References that were only needed for the removed material were deleted.
Changes. First paragraph of Section 1; [C3] reference list. [C121–C123]

Comment M4.

> *Why do the authors present some numerical results in the Appendix?*

Response. We agree that the numerical results belong in the main text. The appendix has been removed. The per-dataset accuracy and AUC values are now reported in Table 3 (p. 19), and the implementation of the two baseline solvers is described in Section 4.2.1. The table of calibrated thresholds and the per-dataset runtime table were removed; the thresholds of every instance are recorded in the supplementary metadata of the reproducibility package.
Changes. Section 4.2.1; [C85–C88] Table 3; [C100] appendix removed. [C118–C120]

Minor issues

Comment m1.

page 2, line 18, what is the "reference point"? It seems to be ambiguous.

Response. We meant that the self-adaptive rule of López et al. was the starting point of our step-size design. The sentence has been removed. The rule is now cited, together with other self-adaptive methods, in the first paragraph of the Introduction.
Changes. Section 1, first paragraph. [C3]

Comment m2.

page 2, line 27, "differing densities" or "different densities"

Response. "Different densities" is the intended meaning. The sentence (the radiotherapy example) has been removed in the shortened Introduction.
Changes. Paragraph removed. [C9]

Comment m3.

page 2, line 35, please polish the sentence "The NSFP is not itself new"

Response. The sentence has been replaced by "Nonlinear SFP and generalized convex feasibility have several relevant antecedents."
Changes. Section 1, third paragraph (p. 1). [C5, C9]

Comment m4.

page 2, line 39, "active index set" or "active set"?

Response. "Active set" is meant. The sentence has been removed. In the algorithm we use the term "approximate active set" (Section 3.2).
Changes. Paragraph removed. [C9]

Comment m5.

page 2, lines 47-49, "We ask instead … make precise" the sentence is difficult for me to understand.

Response. The sentence has been removed. The relation between our function class and those of the closest works is now stated in short sentences in the third paragraph of the Introduction: Li et al. use a convex aggregate proximity function, which componentwise pseudoconvexity does not ensure, and Hu et al. treat quasiconvex inequalities, which include our class.
Changes. Section 1. [C5, C9]

Comment m6.

page 3, lines 48-53, the sentences are too long; please use short sentences.

Response. This paragraph has been removed. Its main point is now stated in one short sentence: "We seek feasibility, rather than minimization of every component."
Changes. Section 1, second paragraph. [C4, C11, C12]

Comment m7.

page 3, the authors present 5 contributions of this paper. Please summarize those main contributions instead of describing too many details.

Response. The five detailed contributions have been replaced by three short items C1–C3.
Changes. Section 1. [C8, C13]

Comment m8.

> *page 5, "Rule (3) has since been extended", please check if it is grammatically correct.*

Response. The sentence belonged to the former Related work section, which has been removed (see M1).
Changes. Removed. [C16]

Comment m9.

> *page 5, line 33, what is the meaning of "drives their proof is for free"? "the nonconvexity sits in the sets"?*

Response. We meant that in the work of Chen et al. the operator is a matrix, so the subgradient inequality used in their proof holds without additional assumptions, and the nonconvexity is only in the constraint sets, not in the map. The passage was unclear and has been removed together with the Related work section.
Changes. Removed. [C16]

Comment m10.

> *page 5, line 35, what is "plain gradient inequality"?*

Response. We meant the usual gradient inequality of a convex function, $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$, as opposed to co-coercivity of the gradient. The passage has been removed.
Changes. Removed. [C16]

Comment m11.

> *page 5, line 44, what is "the closest predecessor"? Do you mean the seminal work?*

Response. We meant the most closely related earlier work (Li et al.), not the seminal work. The passage has been removed. The Introduction now refers to these works as "relevant antecedents".
Changes. Removed; Section 1. [C5, C17]

Comment m12.

> *page 5, line 48, "step" or "step size"?*

Response. "Step size" was meant. The sentence has been removed with the Related work section. [CẦN XÁC NHẬN: trong bản mới vẫn còn "residual-scaled step", "safeguarded diminishing step"; nếu muốn có thể thống nhất cách dùng "step"/"step size" trong toàn bài]
Changes. Removed. [C17]

Comment m13.

> *page 7, line 31, dtap should be DTAP*

Response. Corrected throughout. The acronym is now always written "DTAP".
Changes. All occurrences. [C29, C38, C39]

Comment m14.

> *page 7, line 39, "following the general framework of Combettes 10" should be "following the general framework of Combettes [10]"*

Response. Corrected. The original sentence was removed with the Related work section; the remaining citation now reads "within the framework of Combettes […]" (Section 3.7).
Changes. Section 3.7, first sentence. [C19, C51]

Comment m15.

> *page 9, the subtitles 3.1, 3.2 seem unnecessary. Subsection 3.3 is feature selection, which is suggested to be merged into the Numerical experiments section.*

Response. We agree. The Preliminaries (now Section 2) no longer have subsections. The preliminary subsection on feature selection has been removed, and the feature-selection model (formerly Section 5) is now Section 4.1 of the Numerical experiments.
Changes. Section 2; [C23, C25, C27] Section 4.1. [C61–C80]

Comment m16.

> *page 12, $H := \mathbb{R}^n$ has been stated in the Preliminaries section; why do we repeat $H := \mathbb{R}^n$ once again? Please write the paper as concisely as possible.*

Response. The repetition has been removed from Section 3.1.
Changes. Section 3.1. [C32]

Comment m17.

> *page 13, the subtitle of 4.3, what is the meaning of "standing assumption"?*

Response. We meant assumptions that are in force throughout the analysis. To avoid ambiguity, the subsection and the assumption are now simply titled "Assumptions" (Section 3.3).
Changes. Section 3.3. [C36, C37]

Comment m18.

> *page 18 line 40, [the series] should be $\sum_{k=0}^{\infty} \|x^{k+1} - x^k\|^2 < \infty$. Please follow the advice of the AMS Math formula style as much as possible when presenting the formula.*

Response. Corrected. The series in Proposition 3.11 is now displayed with the limits above and below the summation sign. We also checked the other displayed formulas. [CẦN XÁC NHẬN: đã rà soát các công thức khác theo kiểu AMS chưa? Nếu chưa, bỏ câu cuối]
Changes. Proposition 3.11 (p. 10). [C55]

Comment m19.

> *page 20, line 36, "so no Young splitting is needed". Please polish the writing as much as possible.*

Response. The phrase has been removed from the proof of Corollary 3.13.
Changes. Proof of Corollary 3.13. [C59]

Comment m20.

> *page 20, line 35, $K \cap \bigcap_{i \in I} \mathrm{cvx}$. I don't think this expression is good.*

Response. We now write $\Omega = \{x \in K : F_i(x) \le b_i \text{ for all } i \in I^{\mathrm{cvx}}\}$ in Corollary 3.13, and the analogous inclusion in Stage 1 of the proof of Theorem 3.12 is written in the same way.
Changes. Corollary 3.13; proof of Theorem 3.12. [C56, C59]

General comment.

> *I suggest the authors consider the above comments in their revision. Furthermore, it would be recommended if the authors could make the paper concise.*

Response. We have followed all comments and shortened the paper from 39 to 25 pages (see M1 and M2).

Additional changes made by the authors

While revising the paper we also made the following changes, which were not requested by the referee. We list them for completeness.

1. **Convexity of the fractional criterion.** The submitted version stated that the ratio $F_3(w) = w^\top Qw/\rho^\top w$ is pseudoconvex. We now prove that it is in fact convex on an open neighbourhood of the simplex when $Q \succeq 0$ and $\min_k \rho_k > 0$ (Proposition 4.1). Hence the feature-selection instance belongs to the convex subclass. The assignment of F_3 to the pseudoconvex track used in the experiments remains admissible, and the all-convex assignment is reported as a comparator. This is stated in the abstract, the Introduction, Section 4.1.3 and the Conclusion. [C1, C7, C64, C68, C78, C111]

2. **Domain of the component functions.** The functions are now defined on an open neighbourhood U of K, and (A2) requires continuous differentiability on U. This covers fractional functions whose denominators are positive only near K. A short paragraph after Assumption 1 explains this and why Ω is convex. [C32, C35, C37, C49, C50, C54]

3. **Scope of the method and of the analysis.** New Remarks 1 and 2 state that the proportional weight is not claimed to be unique or optimal, and that the proof uses local bounds on a compact set although no constant is an input of the algorithm. Remark 5 now states precisely that the $O(N^{-1/3})$ bound concerns only the best convex-track violation, and the last stage of the proof of Theorem 3.12 was rewritten more explicitly. [C40, C41, C56, C58]

4. **Experimental protocol.** Section 4.2.2 now describes the protocol exactly: min–max scaling is fitted once on the training pool (not within each selection fold, as stated in the appendix of the submitted version), the voted masks are computed once and reused, the SVM probability calibration uses a fixed seed, and the evaluation is exploratory rather than nested. The formula of the Nogueira stability index and the statistical procedure (exact signed-rank test, Holm adjustment, Friedman test) are given explicitly. [C91–C94] [CẦN XÁC NHẬN: mô tả cũ "fit on each fold's training portion" là sai sót mô tả hay code đã đổi? Response cần nói đúng]

5. **Recomputed values.** The stability values in Table 2, the SVM AUC values in Tables 3 and 4, and the p-values in Table 5 were recomputed with the protocol above; Table 5 now reports raw and Holm-adjusted p-values. Accuracies and all 5-NN results are unchanged. Columns for SG-const and SG-dyn were added. [C99, C100, C102, C104] [CẦN XÁC NHẬN: nêu nguyên nhân chính xác của việc AUC/ổn định thay đổi (seed hiệu chỉnh SVM? hiệu chỉnh $N_f/(N_f-1)$ của chỉ số Nogueira?)]

6. **Revised claims.** Statements of superiority (accuracy, speed, robustness) were removed or qualified. The revised text states that no pairwise comparison is significant after Holm adjustment, that the results show a size–prediction trade-off, and that tuned subgradient methods can match or outperform DTAP. [C1, C95–C98, C101, C103, C106, C111]

7. **New solver diagnostic.** Section 4.4 is new. It reports numerical feasibility witnesses, the activity of the third constraint along the trajectories, a comparison with subgradient methods whose moduli were tuned (including the tuning cost), Movement Libras as an additional instance, and controlled examples with a genuinely nonconvex pseudoconvex constraint (47) (Tables 6–8), together with a reproducibility paragraph. [C57, C71, C105]

8. **Model details.** Implementation details of the feature-selection model are now stated: the regularization constants δ_Q and ε_ρ, the variance floor in the fuzzy relation, the tie-breaking rule in the decision matrix, the correction of the projection-neural-network equation (notation Π and output projection), and the fact that the cumulative selection rule neither enforces a feature budget nor preserves feasibility. [C74, C76, C79, C87]

9. **Smaller corrections.** The cost of the simplex projection is $O(p \log p)$ and the storage of one iteration is $O(nrp + p^2)$; a typo $\{x_k\}$ was corrected to $\{x^k\}$. [C24, C51, C80]

10. **Declarations.** The declaration of generative AI use and the data-availability statement were updated; a reproducibility package now accompanies the paper. [C116, C117]

We hope that the revised manuscript is now suitable for publication, and we thank the referee again for the helpful comments.

Sincerely,
Nguyen Van The, Tran Ngoc Thang and Nguyen Hong Son